Daily Drilling Report
SPD19B-03
9/17/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
52.0	8 1/2"	197.00	72.83	31.70	101.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	264,057.41		12,512,385.63		47.90
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
3,232.00	3,232.00	0.00			

Operations at Report Time
Perform cementing job.

Operations Summary
Perform Wireline Logging job (CBL/VDL/CCL/GR). R/D wireline BHA and sheaves. M/U CTR CMT manifold and L/D. R/U AL-Ghaith 7" Liner handling tools. RIH 7" Liner to 2720m 9 5/8" CSG shoe.

Operations next 24hrs
Cont' RIH 7" Liner on 5 7/8" DP to TD (3232m), Space out and set Liner hanger, release RT. Perform CMT job. POOH hanger RT to surface. Run clean out BHA.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammadreza Asaie	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Behzad Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	9/12/2014	11/12/2014
Medical Drill	9/14/2014	11/14/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/16/2014	11/1/2014
Trip Drill	9/16/2014	11/6/2014

STOP Cards

Number of Reports: 32

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:30	0.50	8 1/2"	J	04	P	P/U and M/U wireline logging BHA (CBL/VDL/CCL/GR) as per below, RIH to 8.11m. (Total Weight: 487 lbs, Tools Max OD: 3.39" , Total Length: 8.11m) Items: Length: Cable Head 0.82m Telemetry 1.553m CCL 1.278m Digital Sonic (CBL) 4.41m Bull Plug 0.045m
00:30	02:00	1.50	8 1/2"	J	04	P	RIH logging BHA on wireline to 9 5/8" CSG shoe. ** Repeat Log f/ 2705m to 2645m.
02:00	05:00	3.00	8 1/2"	J	11	P	Log up f/ 2720.8m (9 5/8" CSG shoe) to surface.
05:00	06:00	1.00	8 1/2"	J	20	P	B/O & L/D logging BHA part by part. R/D wireline sheaves.
06:00	07:00	1.00	Compl...	H	20	P	P/U CTR CMT manifold & keep on rotary. Load pump down plug inside. M/U XOS (5 7/8" XT 57 pin * 5 1/2" FH box) below manifold & XOS (5 7/8" XT 57 box * 5 1/2" FH pin) + FOSV + 5m 5 7/8" pup joint on top of manifold. L/D in one piece.
07:00	07:30	0.50	Compl...	G	20	P	R/U AL-Ghaith 7" liner handling tools. Meanwhile M/U FOSV to X/O (7" VAM TOP pin * 5 7/8" XT 57 box) as kick assembly.
07:30	09:00	1.50	Compl...	G	18	P	HPJSM with all involved crew. Make up shoe track joints as per below & RIH to 65m. Ported float shoe (L-80, 13% Cr) + 2 jts of 7" Liner (29 ppf, L-80 , Vam Top, 13% Cr) + Float Collar (L-80, 13% Cr) + 3 jts 7" Liner (29 ppf, L-80 , Vam Top , 13% Cr) + LDC (L-80 , 13% Cr). *** Check Float shoe & Float collar for holding and passing thru-Ok. *** Apply thread lock on all shoe track connections. *** Apply 12000 ft-lbf TRQ for M/U connections. (Considering baker lock Friction Factor) *** Install 2 Spirolizers on shoe joint, FC joint, joint #54 & 1 Spirolizer on joint #53, 1 Spirolizer on joint #52 as per tally.

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09:00	13:00	4.00	Compl...	G	18	P	RIH 7" CRA, 29 PPF, L-80, 13% Cr, VAM TOP liner f/ 65m to 495m. *** Filled up liner string every 5 joints. *** Applied TRQ: 9400 ft-lbf *** Install #1 7" x 8 1/2" (OD: 8 1/4") Spir- O-lizer in middle of ea. joint around 2 Spir-O-Lok as per tally.
13:00	14:30	1.50	Compl...	G	18	P	Cont. RIH 7" CRA, 26 PPF, SM2535-110, 28% Cr, VAM TOP liner f/ 495m to 651m. *** Filled up liner string every 5 joint. *** Applied TRQ: 10100 ft-lbf *** Install 1# 7" x 8 1/2" (OD: 8 1/4") Spir- O-lizer in middle of ea. joint around 2 Spir-O-Lok as per tally.
14:30	15:00	0.50	Compl...	G	18	P	Change out pipe handling equipment to 5 7/8". P/U & M/U 7" x 9 5/8" CTR Hydraulic set pocket slips liner hanger with 1 pup joint of 7" CRA, 26 PPF, SM2535-110, 28% Cr, VAM TOP on bottom. Install CTR 7" cement Wiper Plug on bottom of slick joint with 5 shear pins. RIH to 658m. *** 7" CTR Liner Hanger assembly components (F/ top to bottom): 1- Liner hanger: 0.91m 2- Lower PBR: 1.85m 3- RCB profile: 0.37m *** CTR Lower PBR specification: Length: 1.85m ID: 7.264" OD: 8.353" *** Total length of 7" liner and liner hanger assy: 659.35m *** Running order is as follows: 1- 1 x 7" CRA,13% Cr, L-80 CTR Ported Float Shoe 2 - 2 x 7" blank CRA,13% Cr, L-80 liner, 29 ppf, VAM TOP 3- 1 x 7" CRA,13% Cr,L-80 CTR Single valve Float Collar 4 - 3 x 7" blank CRA,13% Cr, L-80 liner, 29 ppf, VAM TOP 5 - 1 x 7", CRA, 13% Cr CTR Landing Collar (LDC) 6 -34 x 7" blank CRA,13% Cr, L-80 liner, 29 ppf, VAM TOP 7 -15 x 7" CRA liner, 26 #,SM2535-110, 28% Cr, VAM TOP 8 - 1 x 7" 26#, SM2535-110, 28% Cr, VAM TOP Pup joint 9 - 7" CTR Hydraulic set Liner Hanger Used accessories: 57 x 7"x 8 1/2" Spir-o-lizer centralizer + 114 Spir-o-lok stop collar.
15:00	16:00	1.00	Compl...	G	07	P	Connect TDS and circulate 93 bbls (1.2 string volume) w/ 160 gpm & 100 psi pressure. Check hanger body for any leakage, OK. Record P/U & S/O weights as per below: P/U weight S/O weight W/O Pump: 156 Kips 147 Kips With Pump: 151 Kips 143 Kips Pump Rate Flow Rate SPP 10 SPM 61 GPM 28 Psi 20 SPM 119 GPM 72 Psi
16:00	00:00	8.00	Compl...	G	04	P	Cont. RIH 7" liner string on 5 7/8" DP f/ 658m to 2720m (9 5/8" CSG shoe). *** Fill up string every 10 stds. *** Rabbit all 5 7/8" DP with 3 1/2" drift while RIH. *** RIH speed: 2 min, Slips to Slips

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Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		Compl...				18 Sep, 2014 00:00-00:30 Connect TDS and circulate 123 bbls w/ 160 gpm, 270 psi pressure. Record P/U & S/O weights as follows: P/U S/O With pump: 370 277 kips W/O pump: 384 284 kips Pump Rate Flow Rate SPP 10 SPM 60 GPM 90 Psi 20 SPM 119 GPM 118 Psi 00:30-03:00 Cont. RIH 7" liner string on 5 7/8" DP's f/ 2720m to 3186m, connect TDS and wash down f/ 3186m to 3230m (Btm) w/ 80 gpm & 350 psi pressure. Tag Btm w/8 Kips. *** Fill up string every 10 stds. *** Drifted all 5 7/8" DP to 3 1/2" while RIH. *** RIH speed in open hole: 3 min , Slips to Slips 03:00-03:30 Circulate for 60 bbl w/ 100 gpm, 300 psi meanwhile reciprocate string in 2m interval. 03:30-04:00 POOH and R/B one Std, P/U & M/U 1 jt + 1x 15 ft pup jt (15 ft), drop 1.75" Aluminum ball, R/U CTR Cmt manifold in one piece (3m pup jt + FOSV + X/O + CMT manifold + X/O). Record upward & downward weights (P/U: 443 Kips & S/O: 306 Kips), RIH and tag btm @ 3230m w/ 5 kips, stretch out 0.9m, Space out and position string 2m off btm in neutral point (375kips). R/U CMT lines, L/U to cement unit and STP manifold. 04:00- 05:00 L/U to Rig pumps and chase the ball w/ 160 gpm/ 500 psi. Ball seat after pumping 87 bbls. 05:00-06:00 Pressure up string by rig pump to 2000 psi stepwise and hold pressure for 2 min, S/O weight to 250 Kips and confirm hanger setting. Keep pressure in string and Switch to CMT unit, increase pressure gradually to 3500 psi and hold for 2 min, increase pressure to 4200 psi, ball seat sheared off. P/U string, confirm setting tools is free (PUW=376 kips , SOW=282 Kips). Slack and keep 20 kips on hanger. >>> Meanwhile mix chemicals to prepare 38.5 bbl solution for CMT slurry. *** TOL = 2569m *** Shoe depth = 3028m *** LDC depth = 3163.3m

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
3,232.00	CaCo3/KCL/ Polymer	10.50	53	23.0	21.000	2.7
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.5	0.500	1.600	1.450			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
6	8	0.0	87.0	13.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	10.000	0/100	54.0	1514.0	1940.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7847.0	0.0	0.0			16.74049	0.371

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.20	8.	302	3.0	1,001.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Ahmed 6	Supply Boat			Stand by
Hani	Supply Boat	9/17/2014 20:35	9/17/2014 21:55	Arri. f/Kish,off/L cargo,disem23pax to dd6, tran. 03pax to Borkan, Emb.25Pax.B/L cargo, dep to Kish
Hani	Supply Boat	9/17/2014 12:00	9/17/2014 22:15	Arri f/ SR2, Deli. 01 Bucket of Dope &assisted Mataf Pride for pax transfer, proceeded back to SR2.
Borkan	Supply Boat	9/17/2014 20:00	9/17/2014 21:45	Arrived f/Aban8, Deli. 01 lift car, transferred 02pax and received 03pax f/ Mataf Pride, to ABAN8

Mud Additives

Description	Consumed	Cum Rec	On Loc

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc

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Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
K1		2,732.00	2,586.77
K2		2,851.80	2,696.55
K3		2,901.00	2,740.86
K4		3,029.50	2,856.03
Nar		3,198.50	3,007.23

General Remarks

Comment

>> Requirements:

(Companion flange 7 1/16" x 3" Weco + Tubing hanger R/R tool + Tubing hanger pack-off R/R tool)

>>> Sent:

** Mataf Pride to Kish: (PDK 1 x basket containing full set logging tools + 1 x basket containing 2900m wireline cable & "Cut & Thread" equipment)

>>> Received:

** Moje Darya f/ SR2: (1 x X-mass Tree assy.) + (45 x Spirolizer) + (139 x Spirolock)

** Hani f/ SR2: (1 x bucket Dope for CRA tubular connections)

** Borkan f/ ABAN8: (Petron tool box)

** Mataf Pride f/ Kish: (2 x buckets CRA dope) + (2 x tie back seal nipple + 2 x tie back production packer) + (printer)